Problem Set 28

Definition of the Definite Integral

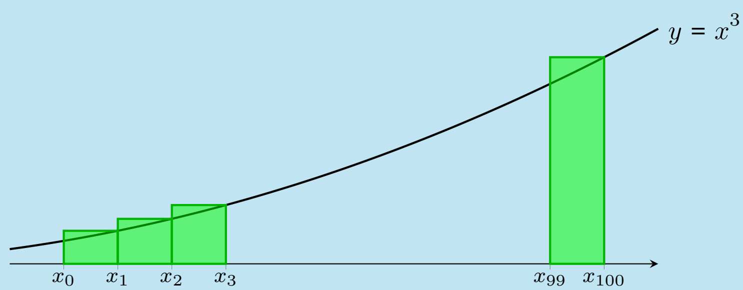
1 Nicholas is approximating the area under a curve using a right-hand sum. The right-hand sum he's written down is

$$\sum_{k=1}^{100} x_k^3 \Delta x \text{ where } \Delta x = \frac{1}{25} \text{ and } x_k = 2 + k\Delta x \tag{1}$$

(a) Fill in the blanks: Nicholas is approximating the area under the curve $y =$ $\underline{\boldsymbol{x^3}}$ from $x =$ $\underline{\mathbf{2}}$ to $x =$ $\underline{\mathbf{6}}$ using a right-hand sum with $\underline{\mathbf{100}}$ rectangles.

Solution

Nicholas's sum can be written out as $x_1^3\Delta x + x_2^3\Delta x + \cdots + x_{100}^3\Delta x$; there are 100 terms, so he is using 100 rectangles. The width of each rectangle is Δx, and the height of the k-th rectangle is x_k^3, so he must be looking at the curve $y = x^3$. His right-hand sum looks like this (with 96 rectangles in the middle not shown):



As we can see from the picture, the area he's approximating goes from $x = x_0$ to $x = x_{100}$. From the formulas $\Delta x = \frac{1}{25}$ and $x_k = 2 + k\Delta x$, we get $x_0 = 2 + 0\Delta x = 2$ and $x_{100} = 2 + 100\Delta x = 2 + 100 \cdot \frac{1}{25} = 6$.

(b) Use Wolfram Alpha (http://www.wolframalpha.com/) to evaluate Nicholas's sum. Note that you'll need to plug the formulas for x_k and Δx in to get something that Wolfram Alpha can evaluate; that is, we can rewrite line (1) as:

$$\begin{aligned} \sum_{k=1}^{100} x_k^3 \Delta x &= \sum_{k=1}^{100} (2 + k\Delta x)^3 \Delta x && \text{(plugging in } x_k = 2 + k\Delta x\text{)} \\ &= \sum_{k=1}^{100} \left(2 + \frac{k}{25}\right)^3 \frac{1}{25} && \text{(plugging in } \Delta x = \tfrac{1}{25}\text{)} \end{aligned}$$

and you can tell Wolfram Alpha to evaluate this by typing

```
sum (2 + k/25)^3 * 1/25, k from 1 to 100
```

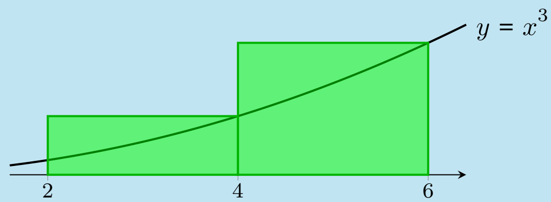
Solution

Wolfram Alpha gives $\dfrac{202608}{625} \approx 324.1728$.

(c) By hand, use a right-hand sum with 2 rectangles to approximate the area Nicholas is interested in.

Solution

We're using a right-hand sum with 2 rectangles to approximate $\int_2^6 x^3\,dx$; here's a picture:



From the picture, we see that $R_2 = 4^3 \cdot 2 + 6^3 \cdot 2 = \boxed{560}$.

If you're wondering how this fits with all of the Δx and x_k notation, here's an explanation: with 2 rectangles, $\Delta x = \frac{6-2}{2} = 2$. We have $x_k = 2 + k\Delta x$, and we could write R_2 in sigma notation as

$$\begin{aligned} R_2 &= \sum_{k=1}^{2} x_k^3 \Delta x \\ &= \sum_{k=1}^{2} (2 + k\Delta x)^3 \Delta x && \text{(since } x_k = 2 + k\Delta x\text{)} \\ &= \sum_{k=1}^{2} (2 + 2k)^3 \cdot 2 && \text{(since } \Delta x = 2\text{)} \\ &= 4^3 \cdot 2 + 6^3 \cdot 2 \end{aligned}$$

(d) Suppose Nicholas wants to use R_n, a right-hand sum with n rectangles (rather than the specific number of rectangles he was using before), to approximate the area he's interested in. Write an expression like that in line (1) that gives R_n. (Your answer should be in terms of n.)

Solution

If we use n rectangles, then $\Delta x = \dfrac{6-2}{n} = \dfrac{4}{n}$, $x_k = 2 + k\Delta x$, and $R_n = x_1^3 \Delta x + x_2^3 \Delta x + \cdots + x_n^3 \Delta x = \sum_{k=1}^{n} x_k^3 \Delta x$.

(e) Use Wolfram Alpha to evaluate your expression for R_n. Please show your work to determine what you enter into Wolfram Alpha (This is basically like (b), except that you're evaluating your answer to (d). Your answer will be in terms of n, since the value of R_n certainly depends on the number of rectangles.)

Solution

We rewrite the sum in (d) as an expression that Wolfram Alpha could evaluate:

$$\begin{aligned}\sum_{k=1}^{n} x_k^3 \Delta x &= \sum_{k=1}^{n} (2 + k\Delta x)^3 \Delta x && \text{(plugging in } x_k = 2 + k\Delta x\text{)} \\ &= \sum_{k=1}^{n} \left(2 + \frac{4k}{n}\right)^3 \cdot \frac{4}{n} && \text{(plugging in } \Delta x = \tfrac{4}{n}\text{)}\end{aligned}$$

We can enter this in Wolfram Alpha as

```
sum (2 + 4*k/n)^3*4/n, k from 1 to n
```

and Wolfram Alpha tells us that the sum is $\boxed{\dfrac{32(10n^2 + 13n + 4)}{n^2}}$.

(f) Is your answer to (e) consistent with your answer to (c)? Explain briefly.

Solution

If we plug $n = 2$ into our answer to (e), we get $R_2 = \dfrac{32(10 \cdot 2^2 + 13 \cdot 2 + 4)}{4} = \dfrac{32 \cdot 70}{4} = 8 \cdot 70 =$ 560, which agrees with our answer to (c).

(g) To find the exact area Nicholas wants, what limit should you take? Evaluate this limit by hand (using your answer to (e)).

Solution

The exact area is $\lim\limits_{n\to\infty} R_n = \lim\limits_{n\to\infty} \dfrac{32(10n^2 + 13n + 4)}{n^2}$. This is an $\frac{\infty}{\infty}$ limit, and it's probably easiest to use relative growth rates to evaluate it: the fastest growing term in the numerator is $32(10n^2)$, so

$$\lim_{n\to\infty} \frac{32(10n^2 + 13n + 4)}{n^2} = \lim_{n\to\infty} \frac{32(10n^2)}{n^2} = \boxed{320}.$$

(h) **Reflect Back.** In this problem, you used the definition of the definite integral as a limit of Riemann sums to evaluate a particular definite integral. In your own words, summarize the process of evaluating a definite integral $\int_a^b f(x)\,dx$ using the definition of the integral. (Which step is hard to do by hand?)

Solution

First, we must write a formula for either R_n or L_n. This formula is a sum, but we need to evaluate it (we don't know how to do this by hand, only using a computer). Then we take the limit as $n \to \infty$ of that formula.

2 Suppose $\int_0^5 f(t)\,dt = 10$. Evaluate four out of the five expressions below. (You do not have enough information to evaluate the remaining one—explain why!)

(a) $\int_0^5 7f(t)\,dt$

Solution

This equals $7\int_0^5 f(t)\,dt = \boxed{70}$.

(b) $\int_0^5 (f(t) + 7)\,dt$

Solution

We can split up the integral as $\int_0^5 f(t)\,dt + \int_0^5 7\,dt = 10 + 7 \cdot 5 = \boxed{45}$.

(We can visualize $\int_0^5 7\,dt$ as a signed area to evaluate it.)

(c) $\int_0^5 f(t)\,dt + 7$

Solution

This is just $10 + 7 = \boxed{17}$.

(d) $\int_0^5 7f(t+7)\,dt$

Solution

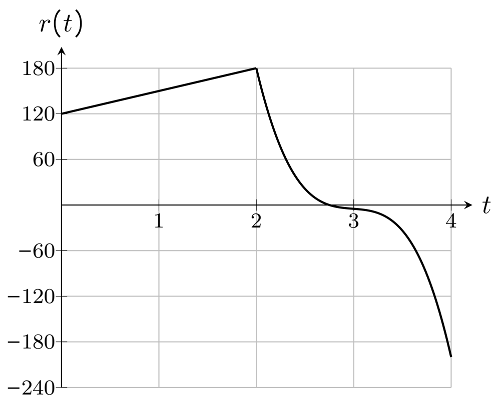
There is not enough information. The graph of $f(t+7)$ is the graph of $f(t)$ shifted left by 7 units, so the behavior of $f(t+7)$ on $[0,5]$ has nothing to do with the behavior of $f(t)$ on $[0,5]$.

(e) $\int_{-7}^{-2} 7f(t+7)\,dt$

Solution

The graph of $f(t+7)$ is the graph of $f(t)$ shifted left by 7 units, so the signed area under $f(t)$ from $t = 0$ to $t = 5$ is the same as the signed area under $f(t+7)$ from $t = -7$ to $t = -2$. That is, $\int_{-7}^{-2} f(t+7)\,dt = \int_0^5 f(t)\,dt$. Therefore, $\int_{-7}^{-2} 7f(t+7)\,dt = 7\int_{-7}^{-2} f(t+7)\,dt = 7\int_0^5 f(t)\,dt = \boxed{70}$.

3 The function $r(t)$ graphed below gives the rate at which water is flowing into or out of a backyard swimming pool one afternoon. (Positive rates mean water is flowing into the pool.) Time t is measured in hours since noon, and the rate $r(t)$ is measured in gallons per hour. At 1 pm, there are 500 gallons of water in the pool.



(a) Write an expression involving a definite integral that gives the exact amount of water in the pool at 3 pm.

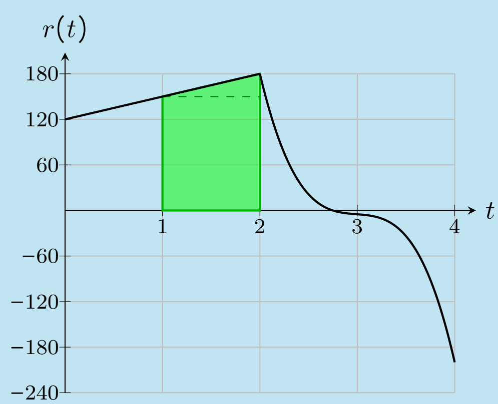
Solution

At 1 pm, there are 500 gallons of water in the pool. The net change in the amount of water in the pool from 1 pm to 3 pm is $\int_1^3 r(t)\,dt$, so the amount of water at 3 pm is $\boxed{500 + \int_1^3 r(t)\,dt}$.

(b) Write an expression involving a definite integral that gives the exact amount of water in the pool at 2 pm, and then evaluate this expression exactly.

Solution

Using the same reasoning as in (a), the exact amount of water in the pool at 2 pm is $500 + \int_1^2 r(t)\,dt$. We can visualize $\int_1^2 r(t)\,dt$ as the area under $r(t)$ from $t = 1$ to $t = 2$; this area forms a trapezoid, and we can find its area by splitting it into a triangle and rectangle:



The rectangle has width 1 and height 150, so its area is 150; the triangle has width 1 and height 30, so its area is 15; therefore, $\int_1^2 r(t)\,dt = 165$, so the amount of water in the pool at 2 pm is $500 + 165 = \boxed{665}$ gallons.

(c) Is water flowing into or out of the pool between 1 pm and 2 pm? Based on that, should there be more water in the pool at 1 pm or at 2 pm? Make sure your answer to (b) is consistent with this.

(Remember that we are told there are 500 gallons of water in the pool at 1 pm.)

Solution

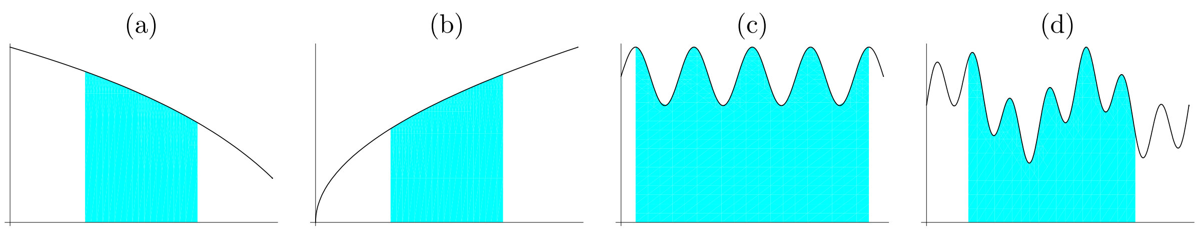
Water is flowing into the pool during this whole time period (because $r(t)$ is positive between $t = 1$ and $t = 2$), so there is more water at 2 pm than at 1 pm. (Our answer to (b) is consistent with this.)

(d) Write an expression involving a definite integral that gives the exact amount of water in the pool x hours after noon.

Solution

Using the same reasoning as in (a), the amount of water in the pool x hours after noon is $\boxed{500 + \int_1^x r(t)\,dt}$. (You might wonder whether this still works for $x < 1$; it does, because $\int_1^x r(t)\,dt = (\text{amount of water } x \text{ hours after noon}) - (\text{amount of water at 1 pm})$, regardless of whether x is greater than or less than 1.)

4 Suppose we use right-hand and left-hand sums to estimate the areas pictured below. Match the pictures with the descriptions below, and briefly explain your choices.



Choices:

A. L_4 will give an underestimate for this area, R_4 an overestimate.

B. L_4 will give an overestimate, R_4 an underestimate.

C. It is very hard to tell from the picture whether L_4 and R_4 will be overestimates or underestimates.

D. Both L_4 and R_4 will give overestimates.

Solution

- Because (a) is a decreasing function, it matches (B).
- Because (b) is an increasing function, it goes with (A).
- It just so happens that (c) has a 5 evenly-spaced peaks, including the endpoints. So, L_4 and R_4 will each give the area of a rectangle whose height it the height of the peak. Hence $L_4 = R_4$ and they are both overestimates. (D) *Note: This answer depends on using exactly 4 rectangles. If we used 3 or 5 rectangles, the answer would be more like (d)–and if we used 8 rectangles, the answer would be close to perfectly accurate!*
- (d) is a typical messy function, going up and down. It is unclear whether Riemann sums are over- or under-estimates without a lot of detailed work. (C)